Video-Informed Scenario Reconstruction for Continuous Robotic Manipulation

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16 October 2026

Abstract

Personal manipulation recordings can inform robot experiments even when they lack calibrated robot-action labels. We study two routes from eight phone recordings to simulated Franka Panda manipulation: scenario reconstruction followed by expert-supervised policy adaptation, and a separate image-based retargeting adapter. Across exploratory comparisons, broader source adaptation improves local manipulation, while goal diversity improves held-out combinations of familiar starts and destinations from 0/12 to 10/12 successes. Neither establishes transfer to fixture loading without loading supervision. Counterbalanced task teaching improves destination selection, but acquisition and skill retention remain distinct failure modes. Two frozen policies subsequently execute feeding followed by loading in one continuous simulated world. In 36 prespecified paired draws, qualified workflow success is 25/36 without assistance (69.4%; exact 95% confidence interval 51.9–83.7%) and 31/36 with controller assistance (86.1%; 70.5–95.3%). Assistance removes several contact failures but introduces a joint-limit violation in one case. A separate online candidate-ranking prototype establishes guarded execution, not improved task performance. The results support bounded simulator capability and identify calibration, acquisition and distribution coverage as the main obstacles to broader transfer.

1 Introduction

Industrial deployment makes transfer between workstations an economic and engineering objective. Humanoid's reported proof of concept with Schaeffler involved picking metallic bearing rings from cluttered bins, transferring them to a buffer table and moving between stations. The account identifies reuse across ring types and autonomous regrasp as intended advantages, with teleoperated data from a physical twin used to adapt a pretrained VLA [1]. This motivates a practical question: can demonstrations teach reusable task structure that reduces additional teaching when the destination, fixture or initial object state changes?

We choose *feeding* and *loading* as a controlled abstraction of this parts-handling workflow. Feeding moves an orange, directionally marked building block into a tabletop staging square; loading transfers it into a shallow fixture. The paired tasks preserve acquisition, transport, orientation and release while changing the destination and placement constraints. They provide a tractable test of skill reuse across workstations, without reproducing bearing-ring geometry or production machinery. The blue numeral distinguishes otherwise similar orientations. Personal phone recordings supply observed configurations and correction examples, rather than measured Panda commands.

These experiments measure bounded layout and goal changes, including preservation of completed work and continuous task execution.

We report three findings about this observation-to-execution bridge. First, robot supervision in reconstructed scenarios supports local learned manipulation, but feeding supervision alone does not teach loading. Second, task-selection errors and grasp-acquisition failures require different remedies. Third, frozen feeding and loading policies execute continuously without a simulator reset, conditional on a narrow scene distribution and externally supplied task switching. Direct visual retargeting and online candidate ranking provide additional comparisons, with negative and inconclusive outcomes retained.

2 Related work

X-VLA uses a flow-matching action model with embodiment-specific soft prompts and projections to accommodate heterogeneous robot data [2]. We adapt its released weights; we do not reproduce foundation-model pretraining. Human video can contribute at different levels: R3M learns reusable visual representations, whereas MimicPlay learns high-level plans from human play and uses robot demonstrations for low-level control [3, 4]. These distinctions motivate separating observations, task intentions and executable motor supervision.

Corrective imitation can address the distribution shift induced by a policy's own actions [5]. Here, expert continuations are admitted only when they execute successfully from the relevant state. They are supervised corrections, not reinforcement-learning updates. Simulation uses robosuite and MuJoCo [6]; simulator truth supports teacher construction and evaluation, with policy access declared separately.

3 Data and system

3.1 Recordings and coordinate conventions

Eight original phone recordings form four synchronized camera pairs: an approximately overhead view and an oblique view. The curated collection contains seven normal feeds, four awkward or pause–resume feeds, three orientation corrections and two already-completed examples. One additional reviewed attempt is excluded from the curated set but retained in the source archive. Fifteen of the 16 curated examples contain sufficient observations for scenario reconstruction.

Claps, common object/hand events and stationary intervals support take boundaries and synchronization. Repositioning between takes is excluded from demonstrations. Classical colour and geometry processing extracts image-space block position, target boundaries and directed numeral orientation, with visibility/confidence masks. The confirmed upright primary view defines orientation; missing or occluded measurements remain unknown. Source timestamps and recording identities connect each derived scenario to its footage.

Target-relative coordinates map to a declared 100 mm virtual square. This is an engineering scale, not a measurement of the physical workspace. The simulation uses a $50 \times 35 \times 30$ mm, 80 g proxy throughout these experiments. No calibrated human-camera depth or physical mass measurement is available.

3.2 Supervision, policy families and controller

The geometric teacher uses simulator object/goal states to generate approach, grasp, lift, transport, orientation, placement and release motions. It supplies robot RGB/action pairs in video-informed scenes. In an initial feasibility comparison, it completes 13/15 reconstructed personal scenarios and 8/15 independently sampled synthetic scenarios. The different start distributions prevent interpreting this as a causal benefit of personal data.

Table 1 distinguishes the principal model families. Early variants compare current observations, dense history, sparse temporal history and corrective continuations. Phase-level and combined-phase chunks, camera-order variants and held-object overlays were prepared; some historical variants also consume a privileged simulator-derived held cue. The dense and sparse catalogues contain 316 and 388 semantic chunks, respectively, with two view orders per chunk. Chunks and camera views are repeated observations of the same source material, not additional demonstrations. A four-second observation history is not necessarily a four-second future action target. The later source-adapted run covers all 439 action windows with 48,000 presentations, balanced at 8,000 per source; equivalent per-update sampling evidence is incomplete for the historical sparse-history run.

Later source-adapted and multi-goal policies use the native interface: two current RGB views, an RGB-derived object crop, proprioception and a task instruction. Predictions specify absolute end-effector position, a 6D rotation representation and gripper command. Each chunk contains 30 actions at 20 Hz, giving a 1.5 s execution horizon before reobservation. An operational-space controller converts these requests into robot motion.

The final loading continuation updates shared vision-language/action parameters and the active prompt/projection rows jointly. It uses AdamW, $\beta = (0.9, 0.95)$, effective batch four, weight decay 0.01 on shared weights, and bfloat16 autocast with float32 weights. Learning rates are 10^{-5} for vision-language weights/prompts and 10^{-4} for the action core/projections. Its 12,000 sample presentations balance feeding and loading, with prior-skill rehearsal and acquisition/preservation examples. This recipe does not implement the published two-stage prompt warm-up schedule.

3.3 Task success and experimental units

We distinguish three measurements. *Supported pickup* requires bilateral contact and at least 10 mm lift for three consecutive control samples; the final workflow audit additionally requires tilt within 10°. *Operational placement* requires containment, appropriate support height, an upright marked face, release and stability. *Strict completion* also requires centre error $\leq$ 15 mm and directed yaw error $\leq 10°$; height tolerance is 2 mm, and loading must be seated. A good yaw score alone is insufficient when the block is tipped.

The primary endpoint of the continuous evaluation is *qualified workflow success*: both stages pass strict completion, no forbidden gripper/table-or-fixture contact is recorded, joint margins remain nonnegative, and any assistance primitive finishes successfully. Contact qualification uses sampled logs, not a certification of continuous-time collision absence.

Exploratory panels reuse source families and inspected scenes. Their trials are not independent human demonstrations; they are reported as finite-panel counts. For goal-diversity comparisons, uncertainty is an exploratory source-family bootstrap over six families. The final reliability panel instead uses 36 prespecified scenario–seed draws; its per-mode success rates receive exact 95% binomial intervals [7]. Paired endpoint-error intervals are exploratory paired bootstraps with 10,000 resamples. None measures variation across independently retrained policies or physical robots.

4 Experiments and findings

4.1 What broader teaching changed

Two early evaluations exposed the limits of the initial adaptations. A 400-trial synthetic feeding/loading panel crossed five model conditions with guarded and unguarded execution. No strict success was recorded; guarded runs were constrained by a 30 s wall-clock cap, making them unsuitable for a clean estimate of the guard's effect. A separate 775-rollout panel evaluated five models on 155 matched feeding scenes, including reconstructed, nearby and independently synthetic starts. All strict passes began already complete. Partial grasps and

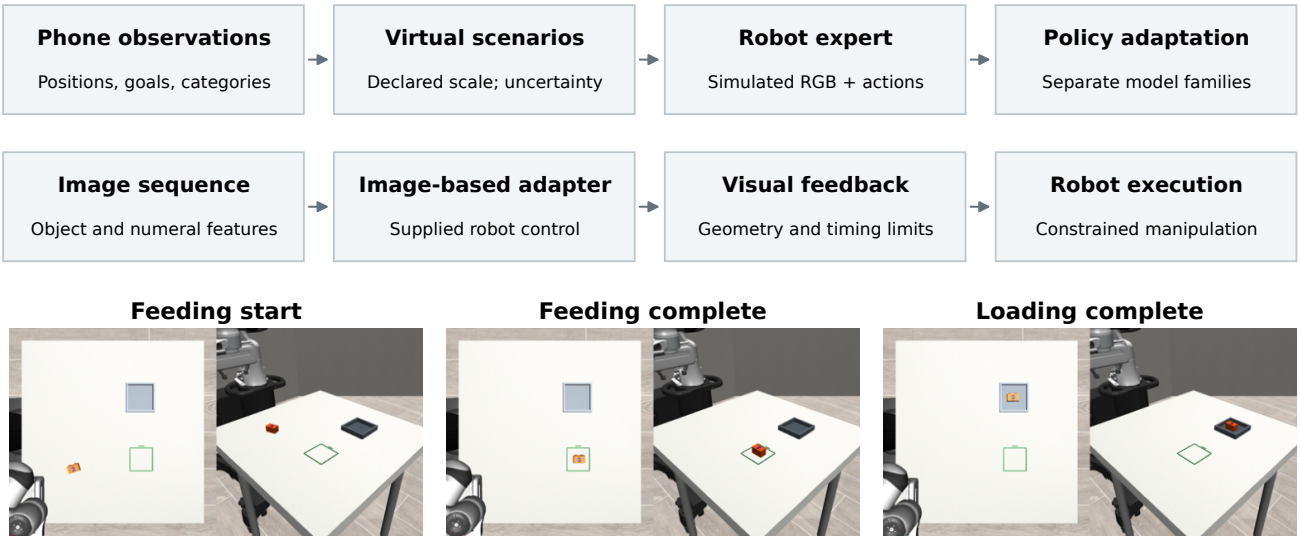


Figure 1: Two implemented routes from personal observations to manipulation. In the learning route, video specifies scenarios and the simulator expert supplies robot-action labels. The visual adapter instead follows image descriptors with supplied robot control and timing assumptions. Below: three original simulator observations from a successful development workflow, showing the feeding start, completed feeding and completed loading. The task switches externally; the simulated world is not reset between stages.

Table 1: Model lineage and the comparisons it supports. These families do not form one uninterrupted training run. Update counts are optimizer steps, not unique demonstrations.

Family	Data and initialization	Training and interpretation
Current-view / dense-history / sparse-history adaptations	Generated robot RGB/actions in reconstructed scenarios; dense and sparse variants add causal history, horizon conditioning and declared held cues.	Separate prompt/action, low-rank and full-weight endpoints. Historical module attribution is incomplete, so these are not a controlled capacity ablation.
Corrective-history adaptation	Continues the sparse-history model using successful expert continuations from policy-visited states.	The evaluated full-weight endpoint adds 500 updates to the sparse endpoint. This is corrective imitation, not PPO or Q-learning.
Source-adapted policy	Released pretrained X-VLA-Pt adapted across all six designated source takes; current-view native interface.	Checkpoints at 3,000 and 12,000 updates; later comparisons use the 12,000-update endpoint.
Fixed-goal / multi-goal / correction-data continuations	All start from the same source-adapted endpoint. Fixed uses six trajectories; multi-goal uses 12; correction-data adds four verified recovery trajectories to six nominal ones.	Each receives 3,000 joint updates, batch four and matched source exposure. Unique trajectory counts differ; one fitting seed per condition.
Loading and task-selection continuations	Further mixed feeding/loading teaching, followed by counterbalanced fixture visibility and acquisition/preservation data.	Controlled 3,000-update continuations separate task coverage, visual shortcuts and retention. Final loading policy is a separate 3,000-update repair continuation.
Continuous-workflow policies	Frozen multi-goal feeding policy plus the final loading policy.	No additional training in the continuous evaluation. Controller assistance is a separate experimental condition.

placements in the corrective-history model therefore did not establish strict completion from an unfinished state.

The subsequent source-adapted policy improves substantially near the recorded configurations (Table 2). It grasps in 31/40 cases, achieves operational placement in 22/40 and strict completion in 17/40, versus zero for the released native-interface baseline. Both fail all 20 broader-start cases. Already-completed controls reveal a separate ability: the adapted policy preserves 8/8 strict endpoints versus 1/8 for the released model. These are conditional comparisons; they do not show that a model with more updates generalizes arbitrarily.

Table 2: Native-interface comparison. Entries are counts; near-source trials reuse six source families. Preservation is measured on already-completed starts and is not a new manipulation success.

Measure	Released	Source-adapted
Near-source pickup	0/40	31/40
Near-source operational placement	0/40	22/40
Near-source strict completion	0/40	17/40
Broader-start strict completion	0/20	0/20
Completed-state preservation	1/8	8/8

A matched continuation comparison isolates useful teaching variation more directly. Fixed-goal and multi-goal variants share an initializer, optimizer budget and source exposure, but use six versus 12 unique teacher trajectories. Multi-goal teaching obtains 10/12 strict successes on withheld combinations of familiar starts and destinations, compared with 0/12 for fixed-goal teaching (Fig. 2). Both remain at 0/12 on fixture loading without loading labels. The full sweep contains 136 baseline, 144 goal/task and 16 recovery episodes; these separate panels are not pooled into one success rate.

Adding four verified recovery trajectories from two source families also fails to solve a quarantined four-state recovery panel: all four compared endpoints score 0/4. Further direct recovery attempts show why teacher qualification matters. An expert continuation can place inside the fixture with 1.22 mm centre error and 0.016° yaw error yet still fail because the block lies on its side. Such a trajectory is not admitted as a successful correction.

4.2 Task selection and grasp acquisition are different failures

Loading-specific supervision enables a new destination. With the same additional 3,000-update budget, a feeding-only continuation remains at 0/4 source-loading and 0/2 withheld task-combination successes. Mixed feeding/loading teaching reaches 3/4 and 2/2, respectively, while strict feeding retention falls from 5/6 to 4/6. Stable in-square feeding placement remains 5/6; the additional failure is a centre-error threshold miss.

Initial task-switching tests show that fixture appearance can dominate the instruction. Counterbalancing visible/hidden fixture images across both tasks raises requested position/footprint attainment from 8/16 to 16/16 and strict completion from 8/16 to 12/16. Strict feeding rises from 4/8 to 8/8; strict loading remains 4/8. In the four remaining loading failures, the block tips during approach, before grasp closure. It is placed at the correct destination but approximately 90° from upright. Delaying release cannot correct an already bad acquisition.

Later acquisition/preservation teaching restores completed-state behaviour but leaves an acquisition trade-off. The final loading policy retains 8/8 completed-state controls while achieving 4/8 on the local loading-initiation panel; an earlier acquisition-focused parent achieves 5/8 but preserves 0/8. In two diagnosed failures, support is lost 1.6 s before opening while the gripper is still commanded closed. Lowering the saved wrist trajectory by 5 mm yields upright lifts of approximately 142 and 140 mm in bounded replay, motivating a controller-level acquisition correction. These replays test pickup, not complete workflows.

4.3 Visibility and language diagnostics

Image resizing and occlusion can remove orientation detail. In one release-state audit, a 512×512 view becomes 224×224, leaving only a tiny visible fragment of the blue numeral; the object crop is absent at that instant. Enlarging the cue changes encoder features more, but a 108-prediction intervention does not remove the approximately 80° wrist-target error or change the opening waypoint. Visibility sensitivity therefore does not establish that the policy uses the numeral correctly in action selection.

Simple instructions also fail as a general motor interface. Across 180 fixed-input predictions from the released model and two historical adaptations, each model produces zero of 18 required paired axis reversals under opposite-direction commands. Later counterbalanced teaching does establish local feeding/loading destination selection, but this is narrower than arbitrary coordinate-language control. The remaining failure cannot be attributed solely to unreadable text or to the controller without these controlled input and trajectory checks.

Replanning and phase supervision. On the corrective-history model, executing ten rather than 30 predicted actions before replanning changes recorded grasps from 2/12 to 0/12, with zero strict successes in either condition. More frequent observation is therefore not sufficient. An offline oracle over 48 proposals improves some approaches, but grasp-feasible states remain 1/6. An observable-phase supervisor is rejected after its guard blocks legitimate grasp attempts; a causal phase predictor also fails its early-release qualification. Completion observers fail their recall gates and receive no authority to advance task phases. These negative diagnostics motivate preserving separate evidence for proposal quality, observer reliability and control constraints.

4.4 Direct visual retargeting: execution without recovered 3D actions

A separate investigation extracts object paths, hand/pinch observations and frozen visual features from the original phone frames without geometric-expert trajectory labels. On 526 sampled images, 440 centre detections pass the confidence threshold and hands are detected in 175 frames; these are coverage measures, not accuracy or grasp recall. Pooled cross-view take retrieval succeeds in only 1/16 cases. Denser sampling recovers additional visible motion, but neither hand landmarks nor planar homographies identify metric lift height.

Calibration sensitivity confirms the ambiguity: all 297 tested intrinsic settings per camera admit a target-corner fit

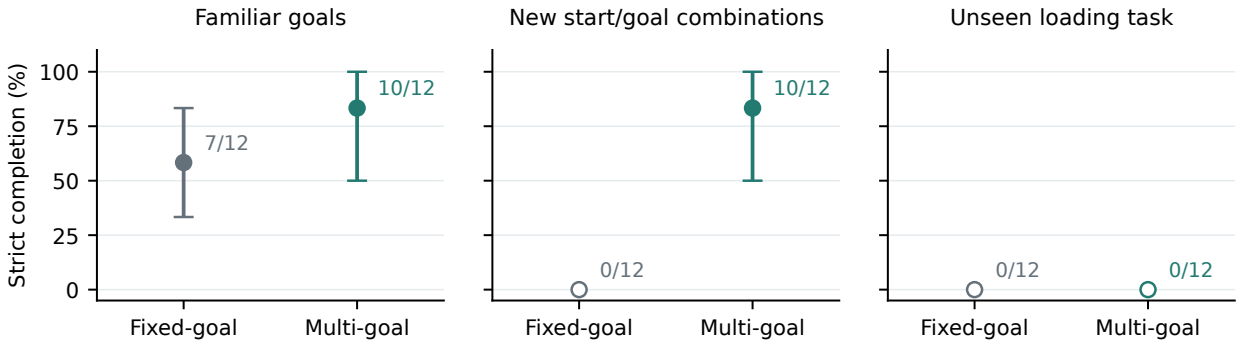


Figure 2: Goal diversity improves new combinations of familiar starts and destinations, but does not teach fixture loading. Each point summarizes 12 trials across six source families. Bars are exploratory 95% source-family percentile-bootstrap intervals (10,000 resamples; fixed fitted models). All-zero cells are shown without an interval because resampling the observed families produces a degenerate interval; this is not certainty of zero population success. Training updates and family exposure are matched, but unique trajectory counts differ.

within the assumed 5 px allowance, while implying substantially different off-plane geometry. A robot adapter therefore uses image descriptors, a supplied virtual scale, calibrated simulator cameras and constrained visual servoing rather than treating image motion as measured XYZ.

On one inspected 7.6 s correction take, five initial sequence/control executions grasp, lift and release, but all miss the task goal by 33.7–35.5 mm. After registering the video endpoint to the fixed task goal, all five conditions—ordered sequence, endpoint-only, temporal shuffle, static reference and exact repeat—pass strict completion. The shared acquisition prefix, fourfold temporal slowdown and supplied release timing remain part of the adapter. A subsequent 736-observation audit finds depth-assumption bias in trajectory-fidelity scores. Thus this is a successful constrained correction example, with no established advantage from intermediate video frames and no general human-to-robot action decoder.

4.5 Continuous feeding followed by loading

The final workflow uses the frozen multi-goal feeding and loading-with-retention policies. A single reset initializes the scene. Each stage receives 1,436 policy actions and 24 settling/scoring ticks at 20 Hz. Switching is externally timed and occurs regardless of success; there is no intermediate reset, homing or object-state injection. An initial four-case development panel succeeds in 4/4 workflows per mode, motivating the larger prespecified evaluation.

The assisted condition changes only loading execution. It applies a frozen 60-action acquisition-height correction while retaining native gripper commands, followed when applicable by a five-action open-gripper clearance primitive. This deterministic assistance is separate from the candidate-ranking service described below. Earlier development tests had already shown a trade-off: an acquisition correction raised loading success from 4/8 to 8/8 but introduced forbidden contacts in 4/8 assisted cases. A broader 40-episode qualification also failed its complete gate. The later evaluation retains these engineering conditions rather than counting endpoint placement alone.

Table 3: Continuous workflow outcomes. Each condition has 36 draws. Qualified success is the intersection of strict task completion and engineering qualification.

Outcome	Native	Assisted
Qualified success	25/36	31/36
95% exact interval (%)	51.9–83.7	70.5–95.3
Both strict stages completed	30/36	31/36
Engineering-qualified	30/36	35/36
Any forbidden contact recorded	6/36	1/36
Negative joint margin	0/36	1/36

The reliability panel samples 36 draws from an equal-probability mixture of two recorded scenes, with independent uniform perturbations of ±5 mm in each tabletop direction and ±10° yaw. Both scenes originate from one camera pair and had been inspected previously. The realized allocation is 22/14. Each draw is executed once in each mode, with counterbalanced order, shared scenario/seed settings and no outcome-based retries. All 72 workflows and 144 stages complete.

Qualified success is 25/36 natively and 31/36 with assistance (Table 3). The observed difference is 16.7 percentage points: 24 pairs succeed in both modes, seven only with assistance, one only natively and four in neither. Five assisted-only wins preserve terminal completion while removing native contact violations; two gain terminal completion. Assistance also loses one native terminal success. The exact marginal intervals are broad, and this panel does not establish superiority across environments or retraining seeds.

Loading errors are heavy-tailed. Median position error is 6.44 mm natively and 4.80 mm assisted, whereas means are 37.88 and 30.21 mm. Median yaw error is 1.05° and 0.48°; means are 5.89° and 5.19°. Maximum errors reach 354.62 mm and 125.23°. Mean paired changes are −7.667 mm and −0.706°, with uncertainty shown in Fig. 3. The position-change interval includes zero; the yaw interval does not, but both are exploratory. Small typical errors coexist with occasional large failures.

The safety trade-off remains material. One assisted workflow has contacts and a minimum logged joint margin of



Figure 3: Continuous-workflow evaluation, retaining all 36 pairs and all failures. (a) Qualified success with marginal exact 95% confidence intervals. (b) Paired outcomes; the outlined draw has an upstream input discrepancy. (c,d) Complete loading endpoint-error distributions, with strict scalar tolerances shown as dashed lines. (e,f) Mean assisted-minus-native loading error with exploratory paired-bootstrap 95% intervals; negative values favour assistance. Error intervals are conditional on the sampled simulator distribution and fixed policies, not calibration or hardware uncertainty.

−0.03106 rad. Across both modes there are 185 forbidden-contact records in seven stages; these are sampled contact entries, not independent collisions. One pair also diverges in upstream feeding images before loading, despite matched random-noise packets. Its loading handover differs, so 35/36 pairs provide exact upstream-controlled loading comparisons. Both outcomes for that draw succeed and remain in every reported denominator.

4.6 Candidate ranking and online execution

A separate supervisory interface collects completed VLA proposals during a 500 ms eligibility window, freezes the candidate set and requests a selection from a multimodal Decisions API. The request includes synchronized camera views, labelled tool-path previews and timestamped simulator metadata. A timely valid choice is rechecked for state freshness and motion guards before execution from the selected plan's correct prefix. Unknown, stale, unsafe or timed-out replies cause an explicit hold/reobserve. Physics continues while waiting; gripper commands are preserved. This is a planning-boundary operation, not an API request on every servo tick.

An earlier asynchronous prototype receives 56 valid replies but activates no selection: 55 abstain and 48 miss its observation-to-choice deadline. Three fallback chunks execute, without strict task success. The revised implementation removes silent first-admissible movement. In a bounded real-model check, its 500 ms window admits one complete plan, not three. Proposal generation takes 479–483 ms, including 178–181 ms of model inference. A mocked abstention produces holds only; one live reply selects the plan after 2.04 s and 27 selected controls execute before the test cap.

These measurements establish integration and explicit timing behaviour, not multi-candidate ranking, smooth sustained motion or improved manipulation.

Offline future-view ranking is a separate privileged diagnostic. Across 12 decisions, a translucent-arm display yields two adequate choices among three selections and nine abstentions; rendered future RGB yields five among five selections and seven abstentions. Conditional accuracy omits abstentions and candidate feasibility, so it does not establish superiority over random selection. Simulator futures are not predictions from a learned world model.

5 Discussion

The experiments identify several mechanisms rather than one global model ranking. Source adaptation improves local grasping and placement; goal diversity expands destination coverage; task-balanced visual teaching removes a local shortcut. None automatically repairs acquisition, skill preservation or unfamiliar starts. A visually correct endpoint also does not establish an acceptable contact history.

The personal-data contribution is concrete but bounded. Recordings define scenes, task categories and visible motion. Expert-generated robot labels enable supervised adaptation; a separate visual adapter can execute one registered correction with substantial supplied control structure. Neither route establishes a general translation from uncalibrated human video to robot actions, and no experiment isolates a causal advantage of personally recorded data over a fully matched synthetic corpus.

The strongest reliability result concerns two related scenes, fixed cameras/materials and frozen checkpoints. Bootstrap and binomial intervals reflect the specified sampling units; thousands of frames do not create thousands of independent tasks. One upstream mismatch limits a clean loading-only treatment interpretation. Simulation parameters, unknown physical geometry and sampled contact logging further limit transfer claims.

Reverse execution remains unresolved: two learned reverse-task attempts fail, and two expert endpoint completions violate contact qualification. Small recovery-data additions do not solve the quarantined states. Additional sequence-alignment studies, matched staged adaptation, reinforcement learning and autonomous phase/reasoning predictors remain unvalidated extensions. The most direct next experiments are independently calibrated recordings, held-out collection sessions, qualified acquisition recoveries and repeated-training comparisons.

6 Reproducibility and availability

The accompanying source package contains the manuscript, plotting code, derived outcome tables and input checksums. Figures regenerate from saved measurements without policy execution. The simulation uses MuJoCo 2.3.7 and robosuite 1.4.1, a 2 ms Euler physics step, Newton solver, and 20 Hz control. Final stability scoring uses 12 samples at 10 Hz over 1.1 s. Evaluation ran on an RTX 4090; inference pauses simulated time in the policy comparisons, unlike the online ranking prototype, so rollout wall time is not a real-time robotics benchmark.

Independent saved-record rescoring reproduces every strict stage decision and workflow qualification in the final panel. Full policy weights, raw recordings and the executable research repository are not publicly released with this manuscript; the included material reproduces the analysis and presentation, not a complete training installation.

7 Conclusion

Video-informed reconstruction and robot-specific supervision produce useful local manipulation and continuous feeding-to-loading execution. Goal diversity and task-balanced teaching improve different aspects of behaviour, while controlled trajectory interventions expose acquisition failures. Controller assistance increases observed qualified success from 69.4% to 86.1% on a narrow paired simulator panel, with remaining contact, joint and generalization limits. Direct visual retargeting and online proposal ranking are promising engineering routes, but their present evidence does not establish general action transfer or reliable autonomous planning.

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